

INDUSTRIAL TRAINING REPORT

At

INTEL TECHNOLOGY SDN BHD

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UNIVERSITI TEKNOLOGI MARA

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At

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Date: 3/2/2022

Name: Ahmad Firdaus Aslam bin Tuan Mohamood

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CHAPTER 1

INTRODUCTION

1.1 Background of Industrial Training



Figure 1: Intel's Headquarters at Santa Clara in 2017

Intel was founded on July 18, 1968, by semiconductor pioneers Gordon Moore (of Moore's law) and Robert Noyce and is associated with the executive leadership and vision of Andrew Grove. Intel was a key component of the rise of Silicon Valley as a high-tech center. Intel Corporation is an American multinational corporation and technology company headquartered in Santa Clara, California. It is the world's largest semiconductor chip manufacturer by revenue and is the developer of the x86 series of microprocessors, the processors found in most personal computers (PCs). Intel supplies microprocessors for computer system manufacturers such as Lenovo, HP, and Dell. Intel also manufactures motherboard chipsets, network interface controllers and integrated circuits, flash memory, graphics chips, embedded processors and other devices related to communications and computing.

Today, Intel is applying their reach, scale, and resources to deliver on bold goals. And they're not doing it alone which they work with their customers and galvanize the industry for positive impact. In partnership with the customers, they advance breakthrough technology and solutions in a variety of segments which is from safe roads and autonomous driving to medical advancements powered by AI & machine learning to technology readiness in education, and more.

1.2 Background of Organization

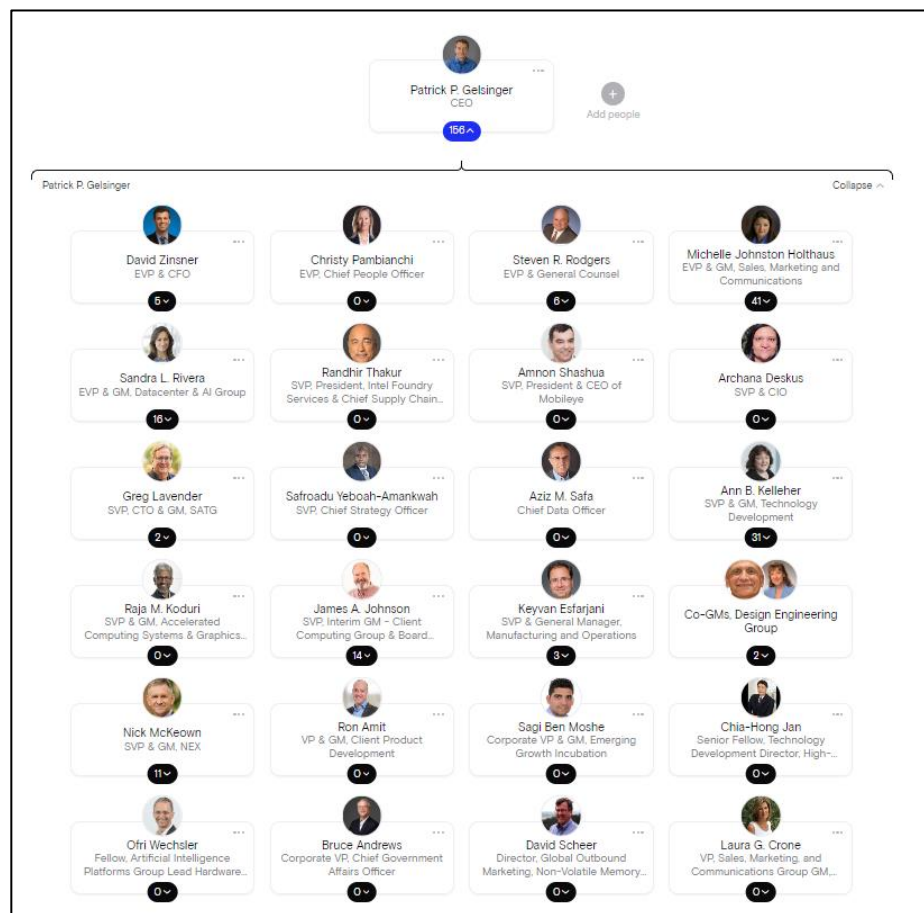


Figure 2: Intel's Executive Team Management

1.3 Background of Department Attached

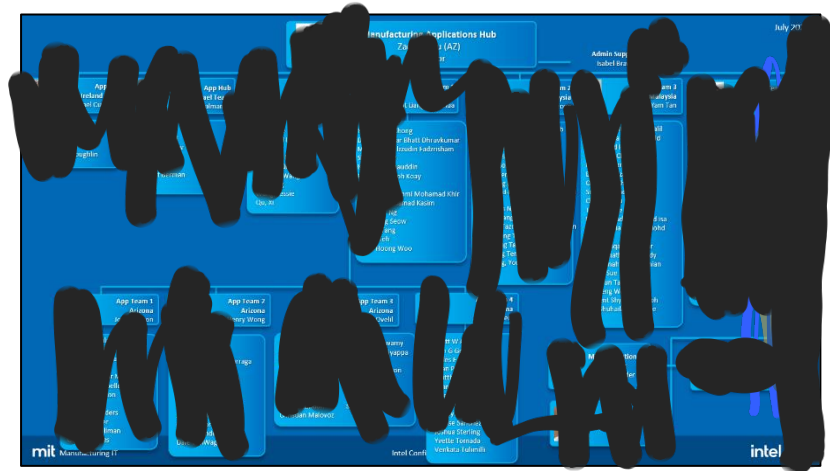


Figure 3: Manufacturing Application Hub Team

I'm currently undergoing my Internship under Manufacturing IT in Manufacturing Execution System (MES). As for me, I'm working as an Automation Engineer Intern under ATM application for Application Malaysia Team 3 Department in Kulim, Kedah supervised by Gim Yam Tan or also known as GY. Figure 3 above show the organization chart in Manufacturing Application Hub Team. Figure 4 below shows the office for our department (KM6) but with the pandemic situation, we are required to work remotely from home, Work From Home (WFH) until an announcement to go back to office being announced.

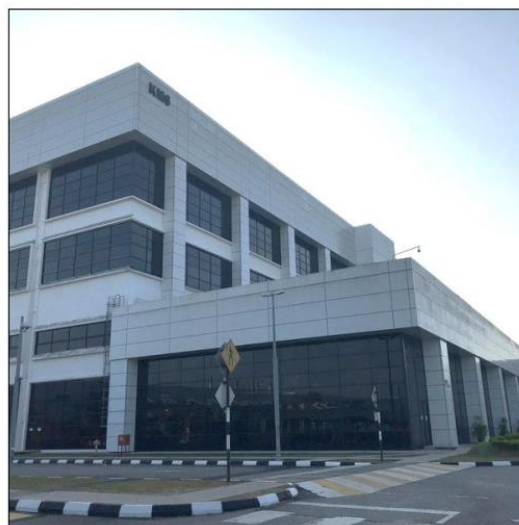


Figure 4: KM6 Department

2.1 Task 1

Fab300 Custom Component Health Check under Excel Macro Visual Basic Application (VBA).

2.1.1 Introduction

The main goal for this project is to convert script from Visual Basic Application (VBA) into PowerShell script. VBA script is using the old script, since the production is using PowerShell most of the time. To optimize the old script it need to convert to PowerShell Scripting.

2.1.2 Implementation

This task is to Smoke test the Fab300 server nodes. In this script there's a lot of duplicates function and variable and a little messy script. For the task, need to convert a script using PowerShell from VBA. Most of result and script is confidential (only can share a few scripts and result).

1. Understand How User Interface works

Figure 5 below shows User Interface (UI) for smoke test. To test it out need to press the "Test Custom Component" button to get the result to check the script is working or not. It will show the result success if the script is works to read from excel through the XML file and vice versa if the script is not working for every attribute that already assigned in the script.

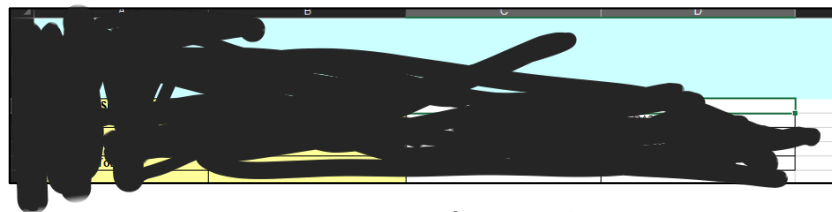


Figure 5: User Interface Smoke Test

2. Check The Script to Understand How it is Works

Figure 6 below shows the script for this task. On this part need to understand line by line to get a better understanding what the commands and functions works.

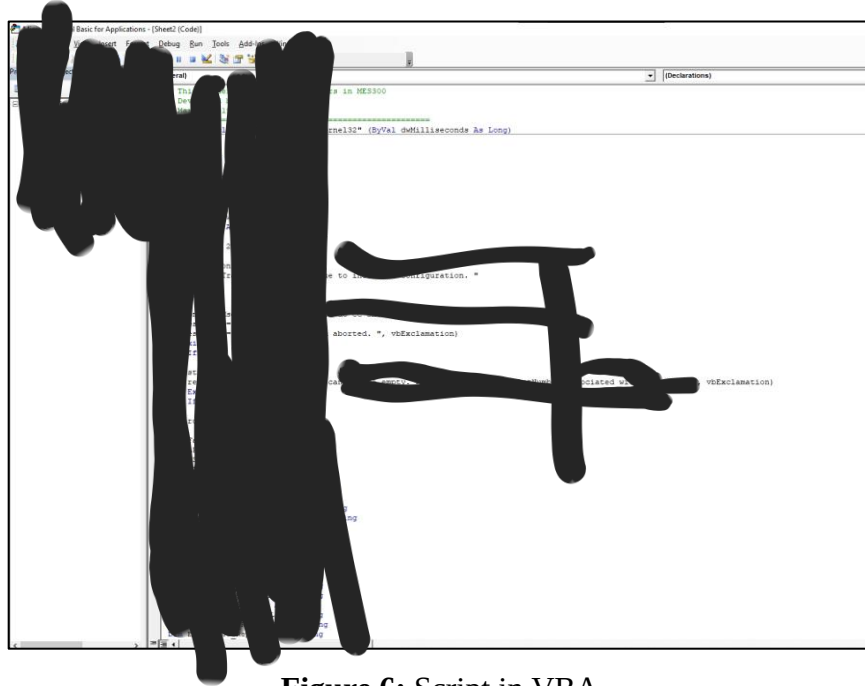


Figure 6: Script in VBA

3. Remove All The Redundant Functions And Variables

Because there are a lot of people who have developed the script, it makes this script a little messy and have few functions and variables that have redundant and duplicates. To ensure there no more duplicates functions and variables, need to rewrite the scripting for efficient script and easy to understand.

4. Convert Script drom VBA into PowerShell

This part is the crucial part because it takes time to invoke the command because There is a lot of differences in terms of syntax for both VBA and PowerShell.

2.1.3 Result and Discussion

After script is already done, then it will be tested in Testing Environment. In this step, we will run the script and discuss on the output. There are a few problems occurred while converting the script which some error cannot be solve because of it too complicated to convert it but this task already pass down to permanent workers to figure it out how to solve it and they already found the solution and ready to be deployed. Figure 7 below shows on how testing being done.

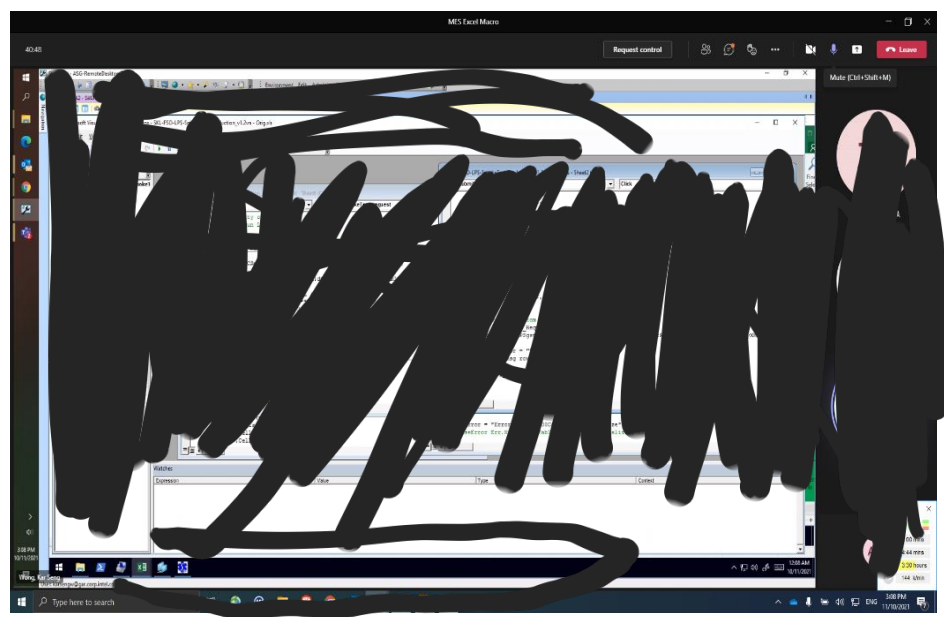


Figure 7: Script testing

2.2 Task 2

Advanced Productivity Family (APF) reporting ICAN script testing.

2.2.1 Introduction

The main goal for this project is to automate some manual tasks or operation to save energy, time and to simplify the current process of manufacturing system and send alert if there any problem occurs while it is running in production server. This project currently used PowerShell.

2.2.2 Implementation

This task has two part which is part one is to schedule to check for saturation. Part two is to enhance script in iCAN to add additional 2 checking before containing reporting server in APF.

1. Recovery Task Scheduler for Saturation (Part 1)

Schedule task to send alert if the saturation if the saturation was breached at certain amount and engage DCT to recover by using automated recovery script. The alert sends through email to notify the server about saturation.

2. Develop Script Based on Pseudocode (Part 2)

In this step, script is being developed based on the requirement gathered within the given period of time. Script will be developed as to ensure the consistency of the scripts to operates very well following the requirements in pseudocode as shown in Figure 8.



Figure 8: One of Pseudocode requirement

2.2.3 Result and Discussion

After the script is completed, it will be tested in the Testing Environment. We will run the script and discuss the results in this phase. If there is a mistake or a new requirement, we will debug and test the script until it meets all of the requirements. This task is already completed after takes a few weeks to complete it and is ready to be deployed in production server.

2.3 Task 3

Sort Web User Interface Health Check exploration.

2.3.1 Introduction

The main goal for this project is to automate website to for automate some manual tasks or operation to save energy, time and to simplify the current process. It can help to do everything from increasing productivity to reducing mistakes and creating a single source of truth for data. Automation can prevent or at least minimize human error if it is set up correctly.

2.3.2 Implementation

This task to automate two websites under Intel which is Solar and Pulsar. Need to do scripting based on attribute in the Hyper Text Markup Language (HTML) tag website to trigger the webpage.

1. Inspect Elements in Website

In this step, need to inspect the elements in HTML attribute to know how to call the attribute such as tag name, id, inner HTML. This part is to call specific attribute to get the object while make an action such as click link, choose dropdown list and etc.

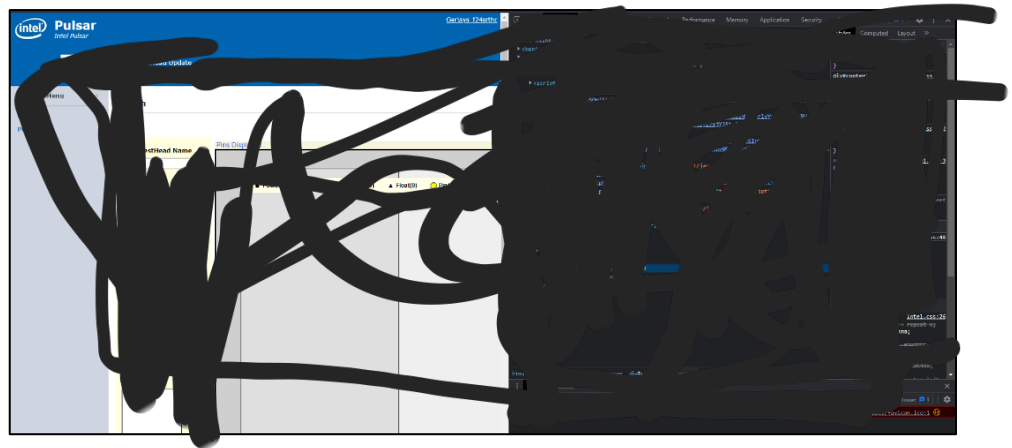


Figure 9: Inspect elements

2. Develop Script

Develop the script based on the requirement needed in this task. Figure 10 below shows the script have been developed to automate the website.

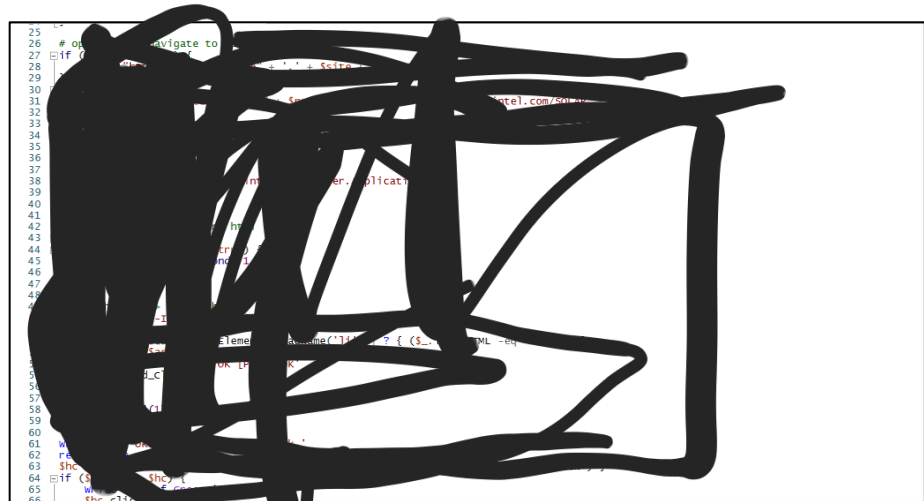


Figure 10: Inspect elements

2.3.3 Result and Discussion

After script is already done, then it will be tested in Testing Environment. In this step, we will run the script and discuss on the output. If there is any error or problem, or any added requirement then we will debug the script and test it again until it fulfils all the requirements. Figure 11 below shows the result that has been successfully working and ready to deploy in the production.

```
OK -I- Select Process
OK [PASS] OK
True
OK -I- Product Alias
OK [PASS] OK
True
OK -I- TestHead Name
OK [PASS] OK
True
OK -I- Pin Displayed
OK [PASS] OK
```

Figure 11: Output automate website

CHAPTER 3

CONCLUSION AND RECOMMENDATION

3.1 Benefit of The Assignments

There are a lot of things that benefits throughout assignment. During first month of internship at Intel, I have been attending many training that has been assigned by my supervisor in order for me to understand their business concept and flow. Below is the list of the training that I have attend throughout my internship:

Table 1: List of training that have been attended

No	Description of tasks/training	Duration
1	New Employee Orientation	2 hours
2	PowerShell Training	2 weeks
3	Manufacturing Process Overview Training	1 week
4	Covid 19 Pandemic Training	1 hour
5	2021 Safety Always Certificate	1 hour
6	New Employee Ethics and Compliance Training 2021	2 hours
7	Privacy Essentials Training	1 hour
8	ISA For System Admin Training	2 hours
9	EHS Orientation	1 hours
10	Intel Malaysia New Hire Curriculum	2 hours
11	Security Orientation Program	1 hour
12	2021 Code of Conduct Training	1 hour

Before supervisor assigned with task, I have been assigned to attend PowerShell training as I did not have any knowledge about PowerShell scripting before, and Intel had provided me this training. From this training, I learned from beginner until advanced module where this module really helps me to complete my scripting task. Not only that, but the training also has assessment at the end of every module which it helps me to test my understanding for all modules before continues next module. Next, I have also been given a chance to learn SQL for PowerShell as an additional knowledge

for me. As part of Intel employee, I'm using this opportunity to explore new things as all the required training are being provided with zero fees.

During my internship under my supervisor, Gim Yam Tan, things that I learned from him is even if we have zero knowledge or not expert in that particular thing, as long as you willing to learn, you will be successful in the future. This is because when I enter Intel, he always keeps in touch with me and inform me whenever there is opportunity to learn and attend any classes that are open for Intel's employees to sharpen our skills in order for us to develop our career. Even though I'm only Intern, but he still offers me the opportunity to learn a new thing.

He also advised me to always grab opportunity as internship is a phase where I need to adapt with working environment and apply my skills and knowledge that I've gained during my study in UiTM. Also, the thing that I learned from him is to always clarify things whenever confusion occur. This is important to avoid wasting our time doing the wrong thing or avoid any miscommunication.

In terms of workers or colleagues, they always keep update me or keep tracking my progress and always be ready to help me whenever I'm facing any problem. This has taught me that teamwork is really important as it will lead to healthy working environment as nobody get left behind. They also taught me that always try to find alternative for every problem, "there must be a better way".

Within other interns, we always set a meeting to discuss on our scripts to solve any problem that occur as we are still learning together during this 6-month period. Not only that, communication skills between the colleagues and maintain a good relationship among them is the most crucial thing to have a good teamwork and holding the responsibility with trust. Without having a good relationship between the colleagues, the task or project will not be able to produce the product of the project with the best quality.

3.2 Summary of Industrial Training

As a conclusion, this Internship training in Intel is a good platform to apply the knowledge that has been gained throughout the studies period as well as learn new skills. The expectations set upon completing this project is to learn in more detail about automation script besides experiencing the real working environment. I've gain valuable work experience during my internship which help me to develop and refine my skills. This may help me to gain more confidence to do the given tasks. By this, I have already understood on the experience of working in the industrial company with a real-life work environment, technology and management as well as the importance of working well in a group. I could also learn how to manage time efficiently to meet datelines and work with professionals in the field.

Other than that, I also learned many problem-solving skills that relates with the scripts. I realize that to become a good worker for the job as well as in the market, we need to have a very good problem-solving skill. By thinking outside the box, it can help us to be more creative and gives us any possible solution for the problem. Next, as a worker that is responsible of the project, we have to obey and follow the timeline and not wasting our time on other thing that does not bring benefits to the company. Other than that, I also learned a lot on improving my communication skills when dealing with the heads and managers. Not only that, I realize that time is gold and by appreciating our times, we will get the wonderful experiences that we cannot get it from anywhere or buy it with our money.

Next, my opinion in order to improve the industrial training in the future is internship students have to force themselves to learn on everything that they managed to learn as they go through their internship program. Besides that, it is expected to learn hands on skills at on site to give myself an edge in the job market. This internship program is the key for a student to learn and understand on the flow of industrial projects so that the students will be more understand and familiar with their working environment one day.

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